Temple Earth / the living cloud - Technical One-Pager

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Evidence layer for ecological sensing: every field reading carries a witnessed, replayable receipt. Non-invasive sensing only - nothing attaches to an organism, nothing enters the habitat, emission lanes are consent-gated (joint guardian + operator approval) and logged. The living system is the participant; computers are scaffolding and verification.

LIVE PRODUCT - REPLAY-VERIFIED COMPUTE API

templeearth.cc - /task /verdict /verify /receipts

- Deterministic LIF simulation over 15 real EM-derived connectome wirings (C. elegans, Drosophila, hemibrain, medulla ~40k neurons, mouse retina, macaque, human segments and more).
- Every task returns task_said + trace_digest; POST /verify replays the task and confirms byte-identical digests. Compute you can check - every task ships its own proof.
- TypeScript port verified bit-identical to the Python reference engine (930/930 checks, Q24 fixed-point, banker's rounding, int64-LE digests, BLAKE2s content addressing).

EDGE / PROTOCOL LAYER

- Receipt chain: content-addressed evidence bundles, quorum witnessing, replay bundles - public ledger at /receipts.
- EcoCell One node: XIAO ESP32-S3 + E22-400M22S Reticulum radio, BME280/BH1750/AD7746, TP4056 charge + MT3608 boost, SMA antenna, consent-gated emit path - manufacturable Gerber/drill/BOM/pick-and-place package complete (Seeed-ready). ~$40 BOM.
- BYOD mode: the same evidence path runs on phones/Raspberry Pis - no custom hardware needed.
- Federation: provenance-preserving adapters for 20+ allied sensor networks (BirdWeather, iNaturalist, Argo, PhenoCam and more).

MEASURED POSITIONING

- Sparse temporal computation benchmarked against dense O(n^2) baselines; honest result: advantage derives from sparsity - biological topology provides identity, provenance and replayable structure. Refused-lane results are published, not hidden.
- License: AGPL-3.0-or-later. Suites green end-to-end: 63/63 protocol e2e, 71/71 live-site e2e, 930/930 compute parity.

TRACTION

- API live in production; verification-audit offers out to Cortical Labs + FinalSpark.
- NSF SBIR/STTR Project Pitch #00131003 submitted; Crowd Supply under review; Elemental Impact application submitted.